SAVEETHA SCHOOL OF ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES COMPUTER SCIENCE AND ENGINEERING

CSA4708 - DEEP LEARNING

ASSIGNMENT 2

CO2 – DEEP FEED-FORWARD AND CONVOLUTIONAL NETWORK

DESIGN Nature: Individual **NAME :** C Prathibha

Duration: 1 Week Total Marks: 50 **REG.NO :** 192424293

CO2: Apply architectural design principles and optimization techniques to construct and train deep feed forward and convolutional neural networks.

Topics: Deep Feed-Forward Networks, Gradient-Based Learning, Back-Propagation, Regularization and Convolutional Networks.

Purpose

Design, train and analyze neural-network architectures rather than merely reproduce a standard model.

Scenario: Smart Agriculture and Crop Disease Detection

An agricultural research centre wants to predict crop yield using rainfall, temperature, soil moisture and fertilizer quantity. It also plans to classify healthy and diseased crop leaves using images. You are required to design, train, optimize and evaluate suitable deep neural network architectures.

TASK 1 – Deep Feed-Forward Network Architecture | 5 Marks

An agricultural centre wants to predict crop yield using rainfall, temperature, soil moisture and fertilizer quantity. A feed-forward network has two hidden layers containing 8 and 4 neurons.

a) **Design the network by identifying the input, hidden and output layers and the number of neurons in each layer.**

The network is designed to predict crop yield using four input features.

- Input Layer: 4 neurons (Rainfall, Temperature, Soil Moisture, Fertilizer Quantity).
- First Hidden Layer: 8 neurons.
- Second Hidden Layer: 4 neurons.
- Output Layer: 1 neuron (Predicted Crop Yield).

Architecture:

Input (4) → Hidden Layer 1 (8) → Hidden Layer 2 (4) → Output (1)

b) Specify the dimensions of the weight matrices and bias vectors.
Using column-vector representation:

Layer	Weight Matrix	Bias Vector
Input to Hidden 1	8 × 4	8 × 1
Hidden 1 to Hidden 2	4 × 8	4 × 1
Hidden 2 to Output	1 × 4	1 × 1

c) Explain how inputs flow through the network to produce a crop-yield prediction.

- The four input values are given to the input layer.
- The first hidden layer processes the inputs using weights and biases.
- The second hidden layer further processes the extracted information.
- The output layer produces the predicted crop yield.

Mathematical representation:

$$Z_1=W_1X+b_1$$
$$A_1=ReLU(Z_1)$$
$$Z_2=W_2A_1+b_2$$
$$A_2=ReLU(Z_2)$$
$$\hat{Y}=W_3A_2+b_3$$

Thus, the network predicts crop yield based on the given agricultural factors

Evidence Expected: Network architecture, neuron counts, weight and bias dimensions, mathematical representation and forward propagation.

TASK 2 – Forward Propagation and Activation Functions | 2 Marks

A neuron receives normalized inputs 0.6, 0.8 and 0.5. Its weights are 0.4, 0.3 and 0.2, and its bias is 0.1.

Given:

Inputs = 0.6, 0.8, 0.5

Weights = 0.4, 0.3, 0.2

Bias = 0.1

a) Calculate the weighted sum received by the neuron.

Formula:

$$Z=(X_1W_1)+(X_2W_2)+(X_3W_3)+b$$

Substitute the values:

$$Z=(0.6x0.4)+(0.8x0.3)+(0.5x0.2)+0.1$$

$$Z=0.24+0.24+0.10+0.10$$
$$\{Z=0.68\}$$

b) Apply ReLU and interpret the output.

ReLU formula:

$$ReLU(x)=\max(0,x)$$
$$ReLU(0.68)=\max(0,0.68)$$
$$\boxed{Output=0.68}$$

Interpretation: Since the weighted sum is positive, ReLU returns 0.68. It removes negative values and keeps positive values.

Evidence Expected: Weighted-sum calculation, activation-function application and interpretation.

TASK 3 – Gradient-Based Learning and Loss Calculation | 6 Marks

A crop-yield network predicts 42 units when the actual yield is 50 units.

a) Explain why a loss function is required and select a suitable loss function for regression.

A loss function measures the difference between the actual output and predicted output. It helps the network understand how much error it has made.

For regression problems, Mean Squared Error (MSE) is a suitable loss function.

b) Calculate the squared error loss using $L = (y - \hat{y})^2$.

Formula:

$$L=(y-\hat{y})^2$$

Substitute the values:

$$L=(50-42)^2$$
$$L=8^2$$
$$\boxed{L=64}$$

c) Explain what the loss value indicates about the prediction.

The loss value is 64. It indicates that the predicted crop yield differs from the actual yield by 8 units. A lower loss indicates better prediction.

d) Describe how gradient-based learning uses the loss to update network parameters.

- The network predicts an output.
- The loss function calculates the error.
- Gradients are calculated using backpropagation.

- Weights and biases are updated using gradient descent.
- This process is repeated to reduce the loss.

Thus, gradient-based learning improves prediction accuracy by minimizing errors.

Evidence Expected: Suitable loss selection, numerical calculation, interpretation and explanation of gradient-based optimization.

TASK 4 – Gradient Descent and Weight Update | 4 Marks

During training, a network weight is 0.5, its loss gradient is 0.8, and the learning rate is

0.1.

a) Calculate the updated weight using the gradient descent rule.

Formula:

$$W_{new}=W_{old}-\eta \times Gradient$$

Substitute:

$$W_{new}=0.5-(0.1\times0.8)$$
$$W_{new}=0.5-0.08$$
$$\boxed{W_{new}=0.42}$$

b) Explain why the update is in the opposite direction to the gradient.

The update is in the opposite direction because gradient descent moves towards the minimum loss.

This helps reduce prediction errors.

c) Discuss the effects of an extremely small or large learning rate.

- Small Learning Rate: Training becomes very slow and takes more time to reach the minimum loss.
- Large Learning Rate: The model may skip the minimum point, fluctuate and become unstable.

Therefore, a suitable learning rate is important for effective training.

Evidence Expected: Gradient descent formula, numerical update, update-direction explanation and learning-rate analysis.

TASK 5 – Back-Propagation Through Multiple Layers | 3 Marks

A crop-yield prediction network contains an input layer, two hidden layers and an output layer. Its prediction differs substantially from the actual yield.

a) Explain how backpropagation determines the contribution of weights to the

prediction error.

Backpropagation is a learning algorithm used to calculate how much each weight

contributes to the prediction error.

Steps:

1. The network produces a predicted output.
2. The loss function calculates the error.
3. The error is propagated backward from the output layer.
4. Gradients are calculated for each weight.
5. Weights are updated to reduce the error.

b) Explain the role of the chain rule in calculating gradients across layers.

The chain rule is used to calculate gradients across multiple layers. It helps determine how changes in one layer affect the final output and loss.

c) Describe how gradients are used to update weights.

Gradients are used in the gradient descent formula:

$$W_{new}=W_{old}-\eta \frac{\partial L}{\partial W}$$

This updates the weights and reduces prediction errors.

Evidence Expected: Backpropagation process, chain-rule explanation, gradient flow and weight updates.

TASK 6 – Activation Functions and Vanishing/Exploding Gradients | 3 Marks A deep network for crop analysis learns slowly because gradients reaching early layers become extremely small.

a) Explain how activation functions can contribute to vanishing gradients.

During backpropagation, gradients pass through multiple layers. Some activation functions produce very small gradients. When multiplied across many layers, gradients become extremely small. This is called the vanishing gradient problem.

b) Compare sigmoid, tanh and ReLU for hidden layers.

Activation	Output Range	Description
Sigmoid	0 to 1	Produces probability-like values but may cause vanishing gradients.
Tanh	-1 to 1	Produces positive and negative values but may also cause vanishing gradients.

ReLU	0 to infinity	Converts negative values to zero and helps reduce vanishing gradients in positive regions.

c) Explain how activation-function choice affects deep-network training.

- Sigmoid may slow down training due to vanishing gradients.
- Tanh performs better than sigmoid in some hidden-layer situations.
- ReLU generally supports faster training in deep networks.

Therefore, ReLU is commonly preferred for hidden layers in deep neural networks.

Evidence Expected: Activation-function comparison, gradient behaviour and impact on training.

TASK 7 – Back-Propagation and Optimization Strategy | 4 Marks

In one experiment, the training loss decreases slowly. In another, it decreases quickly but fluctuates considerably.

a) Explain how the learning rate affects optimization.

The learning rate controls how much the weights change during each training update.

- A small learning rate causes slow learning.
- A large learning rate may cause fluctuations and unstable training.
- A suitable learning rate helps the model reach minimum loss faster.

b) Explain how backpropagation and gradient descent work together in one iteration.

- Forward propagation produces the predicted output.
- The loss function calculates the error.
- Backpropagation calculates gradients.
- Gradient descent updates weights using the gradients.
- The process repeats until the loss becomes smaller.

c) Suggest changes to improve convergence in both experiments.

- Slow loss decrease: Increase the learning rate slightly or use an optimizer like Adam.
- Fast but fluctuating loss: Reduce the learning rate and use a learning-rate scheduler.

d) Explain why the loss curve should be monitored.

The loss curve shows how the model performs during training. It helps identify slow learning, unstable training and overfitting. It also helps determine when to stop training.

Evidence Expected: Relationship between backpropagation and optimization, learning-rate analysis,

suitable adjustments and loss-curve interpretation.

TASK 8 – Regularization and Overfitting Control | 4 Marks

A crop-yield network achieves 99% training accuracy on a crop-category task but only 78% validation accuracy.

a) Identify the likely learning problem and explain the evidence.

The problem is overfitting.

The model has 99% training accuracy but only 78% validation accuracy. This means the model performs well on training data but poorly on unseen data.

b) Explain how L1 and L2 regularization control model complexity.

- **L1 Regularization:** Adds the absolute values of weights to the loss function. It can make some weights zero and helps in feature selection.
- **L2 Regularization:** Adds the squared values of weights to the loss function. It reduces large weights and prevents the model from becoming too complex.

C)Write a regularized loss function using an L2 penalty and explain the regularization parameter.

Formula:

\[L_{total}=L_{original}+\lambda\sum W^2 \]

Where:

- \(L_{original}\) = Original loss.
- \(W\) = Model weights.
- \(\lambda\) = Regularization parameter.

The regularization parameter controls the strength of the penalty. A higher value applies stronger regularization.

e) Explain how dropout may improve generalization.

Dropout randomly disables some neurons during training. This prevents the network from depending too much on particular neurons.

It reduces overfitting and improves performance on unseen data.

Evidence Expected: Identification of overfitting, regularization explanation, loss formulation and dropout-based solution.

TASK 9 – Convolution Operation and Feature Extraction | 3 Marks

An agricultural centre uses leaf images to identify visual signs of disease, including spots, edges and texture.

a) Explain why convolutional layers are suitable for image data.

Convolutional layers are suitable for image data because they automatically identify important features such as edges, spots, shapes and textures.

They help detect disease patterns in crop-leaf images.

b) Demonstrate one convolution operation using a small image matrix and filter, and interpret the result.

Consider the following image matrix:

$$ I= \begin{bmatrix} 1&2&0\\ 0&1&3\\ 2&1&0 \end{bmatrix} $$

Filter:

$$ F= \begin{bmatrix} 1&0\\ 0&-1 \end{bmatrix} $$

Multiply the filter with each 2 × 2 region and add the results.

Output:

$$ O= \begin{bmatrix} 0&-1\\ -1&1 \end{bmatrix} $$

Interpretation: The filter identifies differences in pixel values and helps detect visual patterns such as edges in the image.

c) Explain the roles of filters, stride and padding.

Filters: Detect features such as edges, spots and textures.

Stride: Determines how many steps the filter moves across the image.

Padding: Adds extra pixels around the image to control output size and preserve border information.

Evidence Expected: Convolution-based feature extraction, numerical example and interpretation of filter, stride and padding.

TASK 10 – CNN Architecture and Pooling | 4 Marks

A CNN is proposed to classify leaf images using the architecture: Input Image → Convolution → ReLU → Pooling → Convolution → ReLU → Pooling → Fully Connected Layer → Output.

a) Explain the function of each major component.

Given architecture:

Input Image → Convolution → ReLU → Pooling → Convolution → ReLU → Pooling → Fully Connected Layer → Output.

1. Input Image: Receives the crop-leaf image.
2. Convolution: Extracts important image features.
3. ReLU: Removes negative values and introduces non-linearity.
4. Pooling: Reduces feature-map dimensions.
5. Second Convolution: Extracts more complex features.
6. Second ReLU: Applies activation to the extracted features.
7. Second Pooling: Further reduces the feature-map size.
8. Fully Connected Layer: Uses extracted features for classification.
9. Output Layer: Predicts whether the leaf is healthy or diseased.

b) Explain how pooling reduces feature-map dimensions.

Pooling reduces the size of feature maps while retaining important information.

Example using 2 × 2 max pooling:

\[\begin{bmatrix} 1&3\\ 2&4 \end{bmatrix} \]

The maximum value is 4.

Therefore, the output is:

\[\boxed{4} \]

This reduces the feature-map size and computational cost.

c) Discuss how filter size, stride and number of filters affect feature maps.

- Filter Size: Determines the size of the region being examined.

- Stride: A larger stride reduces the output dimensions.
- Number of Filters: More filters extract more different features and increase the number of output feature maps.

d) Explain how multiple convolutional layers learn increasingly complex features.

- First convolutional layer detects basic features such as edges.
- Second convolutional layer detects shapes and textures.
- Deeper layers identify complex patterns such as disease spots and leaf structures.

Thus, CNN learns features from simple to complex levels.

Evidence Expected: CNN component explanation, pooling operation, feature-map dimensions and hierarchical feature extraction.

TASK 11 – Comparative Evaluation of Feed-Forward and CNN Models | 4 Marks The centre considers Model A, a feed-forward network using numerical weather and soil features, and Model B, a CNN using crop-leaf images.

a) Explain why Model A is appropriate for structured numerical data.

Model A uses a feed-forward neural network. It is suitable for structured numerical data such as rainfall, temperature, soil moisture and fertilizer quantity.

It learns relationships between input values and predicts crop yield.

b) Explain why Model B is suitable for images.

Model B uses a Convolutional Neural Network (CNN). It is suitable for crop-leaf images because it automatically extracts image features such as edges, shapes, spots and textures.

It helps classify healthy and diseased leaves.

c) Compare the models in input representation, feature extraction, parameter organization and computational requirements.

Comparison	Model A – Feed-Forward	Model B – CNN
Input	Numerical data	Image data
Feature Extraction	Learns relationships between numerical features	Automatically extracts image features
Parameters	Weights between fully connected neurons	Filters, kernels and fully connected weights
Computation	Generally lower for small numerical	Generally higher for image processing

	datasets	
Application	Crop-yield prediction	Crop-disease detection

d) Recommend a model for each task and justify the choice.

- Model A – Feed-Forward Network: Recommended for crop-yield prediction because it processes numerical agricultural data effectively.
- Model B – CNN: Recommended for crop-disease detection because it identifies visual patterns from leaf images.

Therefore, both models are suitable for their respective tasks.

Evidence Expected: Technical comparison, relationship between data type and architecture, computational comparison and justified model selection.

TASK 12 – Design Challenge: Optimize and Redesign the Deep Learning System | 8 Marks Changed Requirement

The agricultural centre initially developed a feed-forward network using 10,000 crop records. It must now process 500,000 crop records and 100,000 leaf images. The existing model has high training accuracy, lower validation performance and long training time. Decide whether to retain, modify or redesign the system.

A. Impact Analysis | 2 Marks

a) Explain how the larger numerical dataset and the addition of leaf images affect computational and architectural requirements.

The original system processes 10,000 crop records. Now it must process 500,000 crop records and 100,000 leaf images.

The changes increase:

- Memory requirements.
- Training time.
- Computational power.
- Data processing requirements.

A separate architecture is needed for numerical data and image data.

b) Identify the limitations of applying a feed-forward architecture directly to image data.

1. It requires flattening images into one-dimensional arrays.
2. It may lose spatial information.
3. It requires many parameters.
4. It is less efficient at detecting image features.

Therefore, CNN is more suitable for image processing.

B. Evaluation of the Existing Design | 2 Marks

a) Evaluate the existing model using its training and validation performance.

The existing model shows high training accuracy but low validation accuracy.

This indicates overfitting. The model memorizes training data but does not perform well on new data.

b) Explain whether simply increasing network depth and neuron count is necessarily the best solution.

No. Simply increasing the number of layers and neurons is not always the best solution.

It may increase computational cost and overfitting. Instead, proper architecture design, regularization and optimization techniques should be used.

C. Adaptation or Redesign | 2 Marks

a) Propose a feed-forward network for numerical crop data and a CNN for leaf images.

1. Feed-Forward Network for Crop Data

Architecture:

Input (4) → Dense (64) → ReLU → Dense (32) → ReLU → Output (1)

Inputs:

- Rainfall
- Temperature
- Soil Moisture
- Fertilizer Quantity

Output: Predicted Crop Yield.

2. CNN for Leaf Images

Architecture:

Input Image → Convolution → ReLU → Max Pooling → Convolution → ReLU → Max Pooling → Fully Connected → Output

Output: Healthy or Diseased Leaf.

b) Recommend optimization and regularization methods such as Adam or SGD, learning-rate adjustment, dropout, L2 regularization, batch normalization and suitable activation functions.

The following techniques can improve the system:

1. Adam Optimizer: Helps update weights efficiently.
2. SGD: Updates weights using gradients.
3. Learning Rate Adjustment: Controls the speed of learning.
4. Dropout: Prevents overfitting by randomly disabling neurons.
5. L2 Regularization: Reduces large weights.
6. Batch Normalization: Normalizes layer activations and helps stabilize training.
7. ReLU Activation: Helps improve training speed in hidden layers.

D. Evidence-Based Defence | 2 Marks

a) Design an experiment comparing the original and redesigned systems using loss, validation performance, suitable classification/regression metrics and training time.

Train both systems using the same training and validation datasets.

Compare them using the following measurements:

Evaluation Metric	Purpose
Training Loss	Measures training error
Validation Loss	Measures performance on unseen data
Accuracy	Measures correct classifications
Precision	Measures correct positive predictions
Recall	Measures correctly identified positive cases
F1-Score	Balances precision and recall
MAE	Measures average prediction error
RMSE	Measures prediction error with greater penalty for large errors
Training Time	Measures time required for training

The original and redesigned models can be compared to identify which performs better.

b) Justify whether the original system should be retained, modified or redesigned based on the expected evidence.

I recommend redesigning the existing system.

Reasons:

1. The dataset has increased significantly.
2. Leaf images require a CNN architecture.
3. The existing model suffers from overfitting.
4. Separate networks can handle numerical data and images effectively.
5. Optimization and regularization techniques can improve performance.

Therefore, a feed-forward network for crop-yield prediction and a CNN for crop-disease detection provide a more suitable solution.

Additional Design Considerations

1. Comparison of Optimization Algorithms

Algorithm	Description
SGD	Updates weights using gradients.
Mini-batch SGD	Updates weights using small batches of data.
Momentum	Uses previous gradients to improve learning speed.
Adam	Combines momentum and adaptive learning rates.
RMSProp	Adjusts learning rates using recent squared gradients.

2. Learning Rate Schedule

Start with a learning rate of 0.001. Gradually reduce it during training to achieve stable convergence.

3. Image Augmentation

Use the following techniques to increase image variety:

- Rotation
- Horizontal flipping
- Zooming
- Brightness adjustment
- Cropping

These transformations help the CNN learn better without changing the disease label.

4. Comparison of CNN Architectures

Architecture	Description
LeNet	Simple CNN suitable for small image datasets.
AlexNet	Deeper CNN with multiple convolutional layers.
ResNet	Uses residual connections to train deeper networks.
EfficientNet	Provides good accuracy with efficient computational requirements.

EfficientNet is a suitable option when computational resources are limited and good image-classification performance is required.

Additional design considerations: Compare SGD, mini-batch SGD, momentum, Adam and RMSProp; propose a learning-rate schedule; design label-preserving image augmentation; and compare LeNet, AlexNet, ResNet and EfficientNet for the available image size and computational budget.

Evidence Expected: Analysis of data volume, revised architectures, justified optimization and regularization choices, image augmentation plan and evidence-based evaluation.

Final Mark Distribution – CO2

Task	Topics Covered	Marks
Task 1	Deep Feed-Forward Network Architecture	5
Task 2	Forward Propagation and Activation Functions	2
Task 3	Gradient-Based Learning and Loss Calculation	6
Task 4	Gradient Descent and Weight Update	4
Task 5	Back-Propagation Through Multiple Layers	3
Task 6	Activation Functions and Vanishing/Exploding Gradients	3
Task 7	Back-Propagation and Optimization Strategy	4
Task 8	Regularization and Overfitting Control	4
Task 9	Convolution Operation and Feature Extraction	3
Task 10	CNN Architecture and Pooling	4
Task 11	Comparison of Feed-Forward and CNN Models	4
Sub Total		42
Task 12	Design Challenge – Optimize and Redesign the System	8
TOTAL		50